SAT MATH

Practice tests • Practice Test 1 • Module 2, harder (22 questions, 35 minutes) • Answer key

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Answer key with worked solutions

Directions

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example 3/4 or 0.75); give negative answers with a minus sign; do not use mixed numbers.

Question 1 (Advanced Math)

For $x \neq 4$, which expression is equivalent to $\dfrac{x^2 - 16}{x - 4}$?

(A) $x - 4$

(B) $x + 4$

(C) $x^2 - 4$

(D) $x + \dfrac{4}{x}$

Solution

Answer: (B). $\frac{(x-4)(x+4)}{x-4} = x + 4$.

Question 2 (Algebra)

In the system $kx + 3y = 8$ and $4x + 6y = 5$, k is a constant. For what value of k does the system have no solution?

(A) 4

(B) -2

(C) 8

(D) 2

Solution

Answer: (D). Slopes $-\frac{k}{3}$ and $-\frac{4}{6}$ are equal when $k = 2$; then $2x + 3y = 8$ and $2x + 3y = 2.5$ are parallel.

Question 3 (Advanced Math)

STUDENT-PRODUCED RESPONSE

What is the minimum value of $f(x) = 2(x - 3)^2 + 5$?

Solution

Answer: 5. $(x - 3)^2 \geq 0$, so the minimum is 5, at $x = 3$.

Question 4 (Data Analysis)

In the data set $3, 5, 5, 8, 12$, the value 12 is replaced by 30. Which statement is true?

(A) The mean and the median both increase.
(B) The mean is unchanged and the median increases.
(C) The mean increases and the median is unchanged.
(D) Both are unchanged.

Solution
Answer: (C). The sum grows by 18, so the mean grows by 3.6; the middle value is still 5.

Question 5 (Geometry and Trig)

What is the radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$?

(A) 5
(B) $\sqrt{12}$
(C) 25
(D) 12

Solution
Answer: (A). Complete the squares: $(x-3)^2 + (y+2)^2 = 12 + 9 + 4 = 25$, radius 5.

Question 6 (Algebra) STUDENT-PRODUCED RESPONSE

What is the slope of a line perpendicular to $3x - 4y = 7$?

Solution
Answer: $-\frac{4}{3}$**.** The line has slope $\frac{3}{4}$; the negative reciprocal is $-\frac{4}{3}$ (also -1.333).

Question 7 (Advanced Math)

The graph of $y = x^2 - 6x + c$ touches the x-axis at exactly one point. What is the value of c?

(A) 6
(B) -9
(C) 36
(D) 9

Solution
Answer: (D). One x-intercept means the discriminant is 0: $36 - 4c = 0$, $c = 9$.

Question 8 (Data Analysis)

	Likes math	Does not	Total
Grade 11	30	20	50
Grade 12	25	25	50
Total	55	45	100

A student who likes math is chosen at random. What is the probability that the student is in grade 12?

(A) $\frac{1}{4}$
(B) $\frac{5}{11}$
(C) $\frac{1}{2}$
(D) $\frac{6}{11}$

Solution
Answer: (B). Of the 55 students who like math, 25 are in grade 12: $\frac{25}{55} = \frac{5}{11}$.

Question 9 (Advanced Math)

What is the value of $8^{2/3}$?

(A) $\frac{16}{3}$
(B) 2
(C) 4
(D) 64

Solution
Answer: (C). $\left(\sqrt[3]{8}\right)^2 = 2^2 = 4$.

Question 10 (Geometry and Trig) STUDENT-PRODUCED RESPONSE

Two angles of a triangle measure $47°$ and $68°$. What is the measure, in degrees, of the third angle?

Solution
Answer: 65. $180 - 47 - 68 = 65$.

Question 11 (Advanced Math)

If $3^{x+2} = 81$, what is the value of x?

(A) 2
(B) 4
(C) 25
(D) $\frac{79}{3}$

Solution
Answer: (A). $81 = 3^4$, so $x + 2 = 4$.

Question 12 (Algebra)

What is the solution to $\frac{2}{3}(x - 6) = \frac{1}{2}x + 1$?

(A) 6
(B) 15
(C) -30
(D) 30

Solution
Answer: (D). $\frac{2}{3}x - 4 = \frac{1}{2}x + 1 \Rightarrow \frac{1}{6}x = 5 \Rightarrow x = 30$.

Question 13 (Data Analysis) STUDENT-PRODUCED RESPONSE

A price is decreased by 20%, and the new price is then increased by 25%. The final price is what percent of the original price?

Solution
Answer: 100. $0.8 \times 1.25 = 1$: the final price is 100% of the original.

Question 14 (Geometry and Trig)

If $\sin x^\circ = \cos 40^\circ$ and $0 < x < 90$, what is the value of x?

(A) 40
(B) 50
(C) 140
(D) 45

Solution
Answer: (B). $\sin x^\circ = \cos(90 - x)^\circ$, so $90 - x = 40$ and $x = 50$.

Question 15 (Advanced Math)

The value of a machine, in dollars, t years after purchase is $V = 500(0.92)^t$. Which statement is true?

(A) The value decreases by \$92 each year.
(B) The value decreases by 92% each year.
(C) The value decreases by 8% each year.
(D) The value increases by 8% each year.

Solution
Answer: (C). Each year the value is multiplied by $0.92 = 1 - 0.08$.

Question 16 (Algebra) STUDENT-PRODUCED RESPONSE

If $3x + 2y = 18$ and $2x + 3y = 17$, what is the value of $x - y$?

Solution
Answer: 1. Subtract the equations: $x - y = 1$.

Question 17 (Advanced Math)

The graphs of $y = x^2 - 2x$ and $y = 2x - 3$ intersect at two points. What is the sum of the y-coordinates of these points?

(A) 4
(B) -1
(C) 3
(D) 2

Solution
Answer: (D). $x^2 - 4x + 3 = 0$, $x = 1$ or 3; $y = -1$ and 3; sum 2.

Question 18 (Geometry and Trig)

What is the volume of a cylinder with radius 3 and height 10?

(A) 90π
(B) 30π
(C) 60π
(D) 300π

Solution
Answer: (A). $\pi r^2 h = \pi(9)(10) = 90\pi$.

Question 19 (Data Analysis)

A survey of a random sample estimates that 62% of students walk to school, with a margin of error of 4%. Which is a plausible value of the true percentage?

(A) 56%
(B) 67%
(C) 64%
(D) 70%

Solution
Answer: (C). Plausible values lie from 58% to 66%; only 64% is in that interval.

Question 20 (Advanced Math) STUDENT-PRODUCED RESPONSE

What is the positive solution of $2x^2 - 5x - 3 = 0$?

Solution
Answer: 3. $(2x+1)(x-3) = 0$: $x = -\frac{1}{2}$ or $x = 3$.

Question 21 (Algebra)

For the linear function f, $f(3) = 7$ and $f(7) = 3$. For what value of x is $f(x) = 0$?

(A) 5
(B) 10
(C) -10
(D) 4

Solution
Answer: (B). Slope -1, so $f(x) = -x + 10$ and $f(10) = 0$.

Question 22 (Geometry and Trig) STUDENT-PRODUCED RESPONSE

A circle has radius 9. An arc of the circle has a central angle of $80°$, and its length is $a\pi$. What is the value of a?

Solution

Answer: 4. $\frac{80}{360} \cdot 2\pi(9) = 4\pi$.